

Exercise 1: A random variable X has the following probability function:

Values of X, x:	-2	-1	0	1	2	3
p(x)	0.1	k	0.2	2k	0.3	k

- (i) Find the value of k, and calculate mean and variance.
- (ii) Find $P(X \leq 2)$.
- (iii) Construct the C.D.F. F(X) and draw its graph.

Exercise 2: The following is the distribution function of a discrete random variable X:

x	-3	-1	0	1	2	3	5	8
f(x)	0.10	0.30	0.45	0.5	0.75	0.90	0.95	1.00

- (i) Find the cumulative distribution of X.
- (ii) Find mean and variance.
- (iii) Find $P(X \text{ is even})$ and $P(1 \leq X \leq 8)$.
- (iv) Find $P(X = -3 | X < 0)$ and $P(X \geq 3 | X > 0)$.

Exercise 3: (a) A continuous random variable X follows the probability law:

$$f(x) = Ax^2, 0 \leq x \leq 1$$

Determine A and find the probability that

- (i) X lies between 0.2 and 0.5
- (ii) X is less than 0.3
- (iii) $\frac{1}{4} < X < \frac{1}{2}$.
- (iv) $X > \frac{3}{4}$ given $X > \frac{1}{2}$.

(b) If a random variable X has the density function

$$f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Obtain

- (i) $P(X < 1)$
- (ii) $P(|X| > 1)$
- (iii) $P(2x + 3 > 5)$

Exercise 4: Suppose the life in hours of a certain kind of radio tube has the probability density function:

$$f(x) = \begin{cases} \frac{100}{x^2}, & x \geq 100 \\ 0, & \text{otherwise} \end{cases}$$

Find the C.D.F. of the distribution. What is the probability that none of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation?

What is the probability that all three of the original tubes will have been replaced during the first 150 hours of operation?

Exercise 5: For the following bivariate probability distribution of X and Y, find

- (i) Mean, Variance and Joint distribution of X and Y
- (ii) $P(X \leq 1, Y = 2)$
- (iii) $P(X \leq 1)$
- (iv) $P(Y = 3)$
- (v) $P(Y \leq 3)$
- (vi) $P(X < 3, Y \leq 4)$

Y \ X	1	2	3	4	5	6	p(x)
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
p(y)	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	1

Exercise 6: For the following bivariate probability distribution of X and Y, find:

- (i) Marginal distribution of X and Y
- (ii) Conditional distribution of X given Y=2

Y X	-1	0	1	p(y)
0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{4}{15}$
1	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{6}{15}$
2	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{5}{15}$
p(x)	$\frac{6}{15}$	$\frac{5}{15}$	$\frac{4}{15}$	1

Exercise 7: Suppose that the two-dimensional random variable (X,Y) has joint pdf:

$$f(x,y) = \begin{cases} kx(x-y), & 0 < x < 2, -x < y < x \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Evaluate the constant k
- (ii) Find the marginal pdf of X
- (iii) Find the marginal pdf of Y

Exercise 8: Suppose that the two-dimensional random variable (X,Y) has joint pdf:

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute:

- (i) $P(X > \frac{1}{2})$,
- (ii) $P(Y < X)$
- (iii) $P(Y < \frac{1}{2} | X < \frac{1}{2})$

Exercise 9: For what value of k is $f(x,y) = ke^{-(x+y)}$ a joint pdf of (X,Y) over the region $0 < X < 1, 0 < Y < 1$.

Exercise 10: Suppose that the pdf of (X, Y) is given by:

$$f(x, y) = \begin{cases} e^{-y}, & x > 0, y > x \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Find the marginal pdf of X
- (ii) Find the marginal pdf of Y
- (iii) Evaluate $P(X > 2 | Y < 4)$.

Exercise 11: Suppose that the two-dimensional random variable (X, Y) has joint pdf:

$$f(x, y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find marginal and conditional distributions. Are they independent?

Exercise 12: For the joint density function of a two-dimensional random variable (X, Y) given by:

$$f(x, y) = \begin{cases} \frac{1}{4}, & -1 \leq (x, y) \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find (i) $P(X + Y) \leq a$, where $0 \leq a \leq 1$

(ii) $P(X^2 + Y^2 \leq r^2)$

Exercise 13: If X and Y are two random variables having joint density function :

$$f(x, y) = \frac{1}{8}(b - x - y); 0 < x < 2, 2 < y < 4$$
$$= 0 \quad ; \text{otherwise}$$

Find

- (i) $P(X < 1 \cap Y < 3)$
- (ii) $P(X + Y < 3)$
- (iii) $P(X < 1 | Y < 3)$